Lecture 8: Conscious of the Present; Conscious of the Past: Vision and Memory (cont.)

Introduction to Psychology · Yale University, Spring 2007 · Professor Paul Bloom

Study guide based on the lecture transcript. Source: https://oyc.yale.edu/psychology/psyc-110/lecture-8 (CC BY-NC-SA 3.0).

Before you start

This lecture continues the study of memory: how information gets in and out, why we forget, and how false memories are created. These ideas matter for studying, and for courts that depend on eyewitnesses.

Learning goals

After this lecture you should be able to:

- Compare short-term and long-term memory.
- Describe how meaning and understanding help memory.
- Explain retrieval cues and context-dependent memory.
- Describe the main causes of forgetting, including types of amnesia.
- Explain how suggestion and leading questions can create false memories.

Key ideas

1. Distinctions between Short-Term and Long-Term Memory

- In a real-world change blindness video (Dan Simons), a man giving directions does not notice that the person he is talking to has been swapped. Our perception of reality is much more limited than we think. Usually this is fine, because most things stay the same.
- **Long-term memory** is huge, like a computer's hard drive: words, faces, songs, stories. Nobody knows how big it is, but it is not true that you remember everything.
- **Short-term (working) memory** is small. George Miller said it holds "seven plus or minus two" items (five to nine). Eleven meaningless numbers are very hard to hold.
- The items are **chunks**, units you treat as one thing. "L, A, M, A, I, S, O, N" is eight chunks, but "la maison" is one or two if you know French. Chess masters and architects easily remember meaningful patterns from their field (expertise effects).
- **Maintenance rehearsal** (repeating something, like a phone number, again and again) keeps it in short-term memory, but usually does not get it into long-term memory.

2. How Structure and Organization Affects Long-Term Memory

- In a "depth of processing" experiment, people saw 48 words. Groups judged if each word was in capital letters, if it rhymed with "train," or if it fit the sentence "The girl placed the ____ on the

table." In a surprise test, memory was poor for appearance, better for sound, and best for meaning.

- Memory tricks organize arbitrary facts with vivid images, songs or poems: "The way I find my way around campus is through the hippocampus" (the hippocampus is involved in spatial memory).
- Understanding helps most. A paragraph ("A newspaper is better than a magazine...") is very hard to remember until you learn it is about flying a kite.

3. Memory Retrieval

- **Retrieval cues** are things linked to a memory that bring it back: seeing a cracked window reminds you to replace the windows.
- The **compatibility principle**: you remember better in the context where you learned (context-dependent and state-dependent memory). Scuba divers remembered better when tested in the same place where they learned, on a boat or underwater. Even mood matters.
- **Elaborative rehearsal:** connect new information to many other things (an image, a joke, explaining it to someone). **Elaborative retrieval:** don't give up too soon. Older adults remembered many more high school classmates when pushed with questions about teachers, clubs and sports.
- **Déjà vu** (the feeling that something happened before) is not evidence for psychic powers. It is more common after frontal lobe damage. One theory (from Marcia Johnson): it did happen, half a second ago, but was stored without a time stamp.

4. Memory Failure

- Three explanations of forgetting: **decay** (memory traces in the physical brain fade), **interference** (similar new information blocks old information) and **changes in retrieval cues** (the world changes, so the cues are gone).
- **Childhood amnesia:** it is hard to recover memories from before about age three. Nobody knows why. Ideas: retrieval cues change completely; learning language "reformats" memory; or memory parts of the brain mature later.
- **Retrograde amnesia** ("retro" means past) is loss of past memories. After a blow to the head, memories from just before are lost because they were not yet consolidated (fixed in the brain).
- **Anterograde amnesia** is losing the ability to form new memories, as in Korsakoff's syndrome, the patient H.M., and Clive Wearing (the man in the film from the last class). The damage is to the temporal lobe and hippocampus.
- These patients can still form **implicit memories**. An amnesic gets better each time at filling in a star while looking in a mirror, but always says he has never seen the task. The film "Memento" shows this split.

5. The Power of Suggestion on Memory

- In Steve Ceci's study, a man called Sam Stone visited a class, said "Hi," walked around and left. Some children were told he was clumsy and broke things, and were asked leading questions over several months. Later, some said he ripped a book or messed up a bear.
- These children were not lying; they believed their memories. When about 2,300 experts watched the interviews, most thought the events had happened.

- Adults can also get false memories; some have confessed to crimes they did not commit.
- Elizabeth Loftus showed that small words in questions change eyewitness memory. "Did you see *the* broken headlight?" produced more false memories than "*a* broken headlight." People asked how fast the cars "smashed into" each other later reported broken glass more often than people who heard "hit."

6. Hypnosis, Repressed Memory and Flashbulb Memories

- Hypnosis does not open a hidden store of memories. It makes people eager to please, so they make things up, very believably.
- **Repressed memories:** some adults "recover" memories of childhood sexual abuse in therapy. Some psychologists think some of these memories are real; most think they cannot be trusted and may be created by the therapist. Whether memories are ever repressed is controversial; that false memories can be implanted is not.
- **Flashbulb memories**, like where you were on September 11, 2001, feel vivid but cannot be trusted; retelling changes them.
- "Being sure is no guarantee that a memory isn't false, reconstructed or even implanted."
- Three morals: there are many types of memory; the key to remembering is organization and understanding; and you can't trust some of your memories.

7. Question and Answer on Memory

Knowing how to play the piano is long-term and implicit memory: "your fingers know and not your mind." Bloom doubts that true photographic memory exists.

Key vocabulary

Term	Meaning in simple English	Example from the lecture
long-term memory	the large, lasting memory store	60,000-80,000 English words, faces, songs
short-term (working) memory	small memory for what is in your mind right now	"seven plus or minus two" items
chunk	a piece of information you treat as one unit	"la maison" instead of eight letters
depth of processing	how deeply you think about information	meaning is remembered better than appearance
retrieval cue	something that helps bring a memory back	a cracked window
context-dependent memory	better memory in the same place or state where you learned	divers on a boat or underwater
childhood amnesia	difficulty remembering events from before about age three	people's "first memories"
retrograde amnesia	loss of memories from the past	after a blow to the head

Term	Meaning in simple English	Example from the lecture
anterograde amnesia	loss of the ability to form new memories	H.M., Clive Wearing
leading question	a question that suggests a certain answer	"Did you see the broken headlight?"
flashbulb memory	a very vivid memory of an important event	where you were on 9/11

Check your understanding

1. According to George Miller, how much can short-term memory hold? What is a chunk?
2. Why can a chess master remember a chessboard after a short look, while a beginner cannot?
3. In the depth of processing experiment, which group remembered the most words? What does this show?
4. You learned this course in one lecture hall. Where should you take the final exam? Why?
5. Give three explanations of why we forget.
6. Compare retrograde and anterograde amnesia.
7. How did some children in the Sam Stone study come to "remember" things that never happened?
8. A friend says, "I am completely sure about my memory of 9/11, so it must be true." How would Bloom answer?

Answer guide

1. About seven items, plus or minus two. A chunk is a unit of information you treat as one thing.
2. If the position makes sense in chess, the master understands it and stores it as a few chunks. For the beginner, every piece is separate.
3. The group that judged meaning. The deeper you think about something, the better you remember it.
4. In the same hall. The room gives retrieval cues that help bring the memories back.
5. Decay, interference, and changes in retrieval cues.
6. Retrograde amnesia is loss of past memories. Anterograde amnesia is losing the ability to form new memories.
7. They heard that Sam Stone was clumsy and were asked leading questions over several months. They were not lying; they believed it.
8. Being sure is no guarantee. Retelling changes flashbulb memories, and people can be completely sure of false memories.

English practice

Useful phrases from the lecture

- "The key to remembering is organization and understanding." - pattern *The key to X is Y* names the most important factor.
- "The more you understand something, the easier you'll remember it." - pattern *The more..., the...*
- "It turns out that you'll remember it better if..." - introduces a research finding.
- "Nobody really knows." - admits that science has no answer yet.
- "Being sure is no guarantee that..." - warns that a feeling is not proof.

Speaking task (with your AI tutor)

- Explain in 4-6 sentences how to study for an exam, using this lecture. Use: *depth of processing*, *retrieval cue*, *elaborative rehearsal*.
- Explain in 4-6 sentences how a false memory can be created. Use: *leading question*, *suggestion*, *eyewitness*.

Writing task

- Should courts fully trust eyewitnesses who are sure about what they saw? Write one paragraph (80-120 words) using two studies from the lecture (Ceci and Loftus).

Study with your tutor

Things to say to your tutor in Live conversation:

- "Let's review Lecture 8. Ask me questions about short-term and long-term memory, one at a time."

- "Quiz me on the key vocabulary of Lecture 8."
- "I will explain the Sam Stone study in my own words. Check my understanding and my English."

Readings from the original course

- Gray, Peter. Psychology (5th edition), chapters 8 and 9
- Hoffman, Donald D. "Visual Intelligence: How We Create What We See." In The Norton Psychology Reader. Edited by Gary Marcus. New York: W. W. Norton & Company, 2006. pp. 81-90
- Hughes, Howard C. "Sensory Exotica: A World Beyond Human Experience." In The Norton Psychology Reader. Edited by Gary Marcus. New York: W. W. Norton & Company, 2006. pp. 91-95
- Schacter, Daniel L. "Searching for Memory: The Brain, the Mind, and the Past." In The Norton Psychology Reader. Edited by Gary Marcus. New York: W. W. Norton & Company, 2006. pp. 165-176

My notes